DIMENSION OF FIBRES OF A MORPHISM OF LOCALLY ALGEBRAIC SCHEMES: BOUNDS, OPENNESS AND THE GENERIC VALUE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof states that the author will freely use the results of the dimension theory lemma for X, Y and the fibres; the sentence is reproduced verbatim and those results are navigation stubs, while the computation of d is written out inside the counted span.

Editorial note. The counted spans carry 6 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-dimension-fibres-locally-algebraic is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Transcendence degree description of the dimension of a scheme and of its fibres. Same chapter, Lemma lemma-dimension-locally-algebraic, used freely at proof lines 3979-3981 and in the computation at 3986-3990. Supplies the identification of d with the fibre dimension at the generic point.
- Openness of the locus where the fibre dimension is bounded. Morphisms, Lemma morphisms-lemma-openness-bounded-dimension-fibres, invoked at proof lines 3991-3992. Supplies the lower bound and the density of equality.
- Fibre dimension bounds in codimension one and in higher codimension. Same chapter, Lemmas lemma-dimension-fibre-in-codim-1 and lemma-dimension-fibre-in-higher-codimension, invoked at proof lines 3993-3995. Supplies parts (2)(c) and (2)(d).
- Locally finite type is preserved by the morphism. Morphisms, Lemma morphisms-lemma-permanence-finite-type, invoked at proof lines 3975-3978. Supplies that the fibres are locally algebraic over the residue field.

Source reference. Original lemma 0B2I: sources/stacks-varieties/pinned-original-source.tex, source lines 3597–3597. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0B2K: sources/stacks-varieties/pinned-original-source.tex, source lines 3671–3671. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 0A21: sources/stacks-varieties/pinned-original-source.tex, source lines 3783–3783. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02FZ: sources/stacks-chow/tags.txt, source lines 3210–3210. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01T8: sources/stacks-chow/tags.txt, source lines 2365–2365. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let k be a field. Let $f : X \to Y$ be a morphism of locally algebraic k-schemes.*

1. *For $y \in Y$, the fibre X_y is a locally algebraic scheme over $\kappa(y)$ hence all the results of Lemma 0A21 apply.*
2. *Assume X is irreducible. Set $Z = \overline{f(X)}$ and $d = \dim(X) - \dim(Z)$. Then*
   (a) *$\dim_x(X_{f(x)}) \geq d$ for all $x \in X$,*
   (b) *the set of $x \in X$ with $\dim_x(X_{f(x)}) = d$ is dense open,*
   (c) *if $\dim(\mathcal{O}_{Z,f(x)}) \geq 1$, then $\dim_x(X_{f(x)}) \leq d + \dim(\mathcal{O}_{Z,f(x)}) - 1$,*
   (d) *if $\dim(\mathcal{O}_{Z,f(x)}) = 1$, then $\dim_x(X_{f(x)}) = d$,*
3. *For $x \in X$ with $y = f(x)$ we have $\dim_x(X_y) \geq \dim_x(X) - \dim_y(Y)$.*

Proof. The morphism f is locally of finite type by Morphisms, Lemma 01T8. Hence the base change $X_y \to \mathrm{Spec}(\kappa(y))$ is locally of finite type. This proves (1). In the rest of the proof we will freely use the results of Lemma 0A21 for X, Y, and the fibres of f.

Proof of (2). Let $\eta \in X$ be the generic point and set $\xi = f(\eta)$. Then $Z = \overline{\{\xi\}}$. Hence

$$d = \dim(X) - \dim(Z) = \mathrm{trdeg}_k\kappa(\eta) - \mathrm{trdeg}_k\kappa(\xi) = \mathrm{trdeg}_{\kappa(\xi)}\kappa(\eta) = \dim_\eta(X_\xi)$$

Thus parts (2)(a) and (2)(b) follow from Morphisms, Lemma 02FZ. Parts (2)(c) and (2)(d) follow from Lemmas 0B2K and 0B2I.

Proof of (3). Let $x \in X$. Let $X' \subset X$ be an irreducible component of X passing through x of dimension $\dim_x(X)$. Then (2) implies that $\dim_x(X_y) \geq \dim(X') - \dim(Z')$ where $Z' \subset Y$ is the closure of the image of X'. This proves (3). □